

WRF Tools case report

Case

Item	Result
Wrfout File	wrfout_d01_2006-07-02_00_00_00.nc
Grid Id	1
Dimensions	{"Time": 25, "south_north": 109, "west_east": 109, "bottom_top": 50, "bottom_top_stag": 51, "soil_layers_stag": 4, "west_east_stag": 110, "south_north_stag": 110}
Selected Point	{"latitude": 54.011009216308594, "longitude": 6.60699462890625, "x": 54, "y": 54}

Quality Control

Item	Result
Errors	1
Warnings	0
Issues	[{"severity": "ERROR", "variable": "QVAPOR", "check": "physical-range", "message": "5 values outside [0.0, 0.05] kg kg-1; first value -5.04646e-12 at (4, 22, 22, 57)", "count": 5, "first_index": [4, 22, 22, 57], "firs...

Surface Wind

Item	Result
Mean 10M Wind Speed M S	5.702756404876709
Maximum 10M Wind Speed M S	6.628946781158447
Turbulence Intensity U10 Percent	10.344138148956008

Boundary Layer

Item	Result
Mean Wrf Pblh M	59.371002197265625
Mean Wrf Pblh Excluding Zero M	61.84478759765625
Mean Diagnosed Pblh M	40.80378807801753
Mean Convective Velocity Scale M S	0.0
Stability Class Counts	{"neutral": 1, "stable": 10, "very stable": 14}
Diagnostic Method	First model level at least 0.5 K warmer than the lowest level.
Interpretation	This discrete threshold estimate is constrained by model-level spacing and is not equivalent to WRF PBLH. Negative surface heat flux and predominantly stable classes support a shallow stable boundary layer.

Wind Energy

Item	Result
Rotor Bottom M Agl	40.0
Rotor Top M Agl	200.0
Mean Rotor Equivalent Speed M S	8.94480563306793
Mean Shear Exponent	0.13661014765910726
Mean Veer Degrees	21.29163088260664
Weibull Shape	12.977150713681887
Weibull Scale M S	5.9341242827333875
Mean Wind Power Density W M2	116.9664535522461

Spectral And Coherence

Item	Result
U10 Fitted Log Slope	-1.1285758856503003
Integral Time Scale Hours	1.884550366050926
Integral Length Scale M	38689.67401311479
Mean Lateral U10 Coherence	0.7423289666389511
Record Length Warning	Only 25 hourly samples: spectra demonstrate the API but cannot resolve an atmospheric inertial subrange.

Thermodynamics

Item	Result
Surface Relative Humidity Percent	81.14533233642578
Maximum Equivalent Potential Temperature K	419.0875244140625
Maximum Brunt Vaisala Frequency S 1	0.06772540509700775
Inversion Level Count First Time	11

Turbulence And Flow

Item	Result
Mean Resolved Tke M2 S2	0.9660675525665283
Mean Shear Production M2 S3	5.536619937629439e-06
Mean Buoyancy Production M2 S3	8.404942491324618e-06
Maximum Absolute Vorticity S 1	4.020918640890159e-05
Low Level Jet Detected First Time	True
Jet Core Speed M S	11.65857219696045
Jet Core Height M Agl	12398.648171317764

Precipitation

Item	Result
Domain Mean Total Mm	1.936484694480896
Domain Maximum Interval Mm	19.69297218322754

Terrain Landuse Resolution

Item	Result
Maximum Terrain Slope Degrees	2.171964594472876
Landuse Category Statistics	{"1": {"count": 1295, "mean": 285.83861065131816, "std": 1.9842198731448542, "minimum": 279.9344787597656, "maximum": 296.2147521972656}, "2": {"count": 93, "mean": 290.99849545058385, "std": 1.95717966640713, "minimu...
Coarsen Regrid Comparison	{"bias": 0.05678467102992682, "mae": 0.2214457579928007, "rmse": 0.34022757023090217, "correlation": 0.9964989893636517}
Temperature Integral Conservation	{"source_integral": 2452295866384384.0, "target_integral": 2452295866384384.0, "relative_error": 0.0}
Boundary Difference K	-0.36881053062199953
Vector Rotation Speed Max Error	1.9073486328125e-06

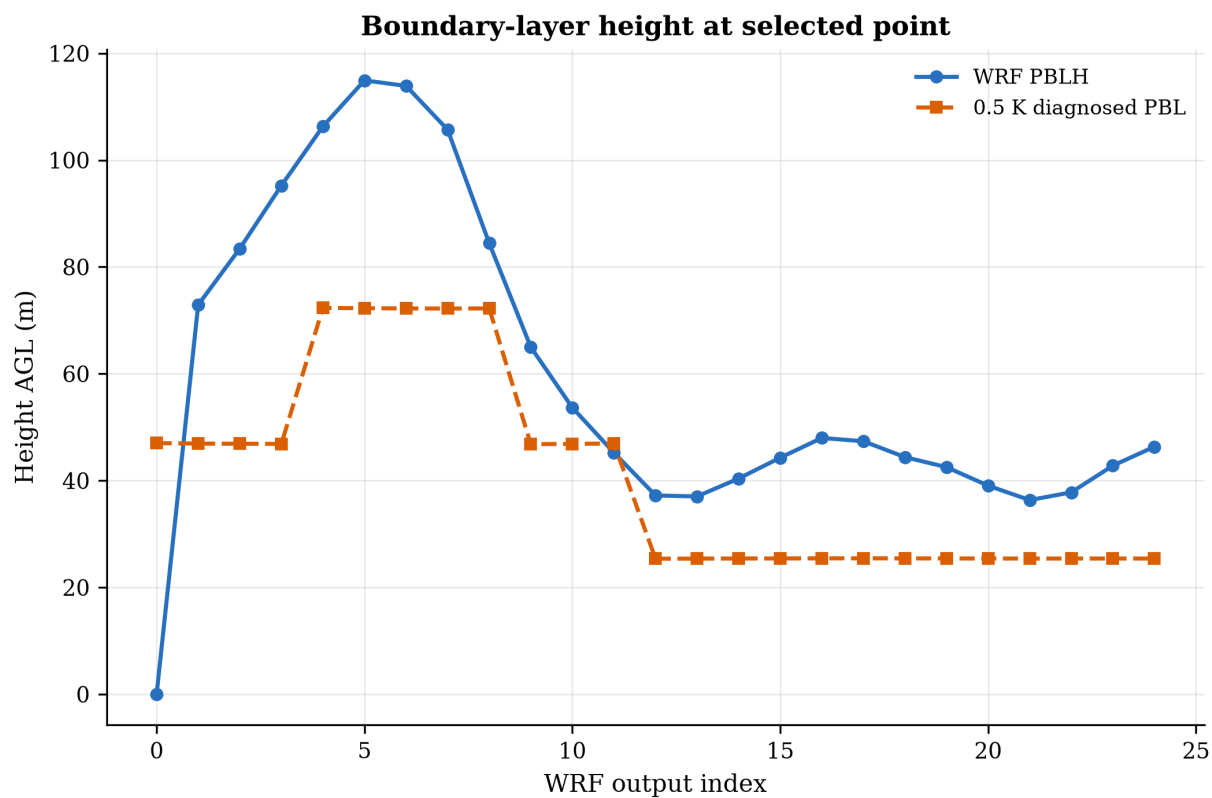
Temporal

Item	Result
Lag1 Persistence	0.8388461666948307
Ramp Threshold M S	0.6124426345834966
Ramp Event Count	1
Minimum Diurnal Mean Speed M S	4.8861236572265625
Maximum Diurnal Mean Speed M S	6.628946781158447
Peak Diurnal Hour	5

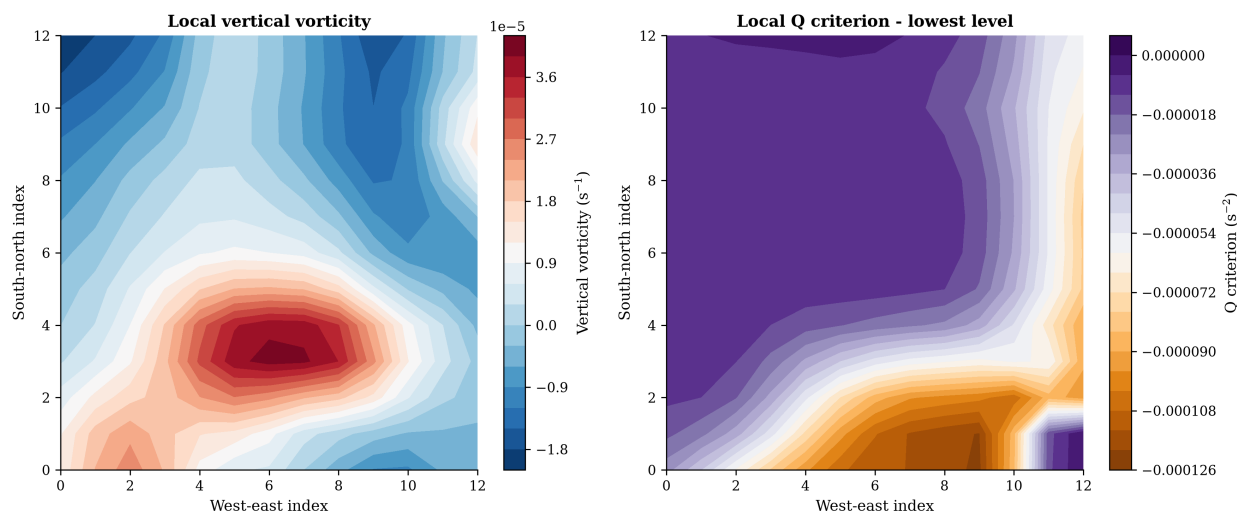
Openfast Conversion

Item	Result
Bts Roundtrip Max Error M S	6.686200431715861e-05
Wnd Roundtrip Max Error M S	0.0010901477125866421

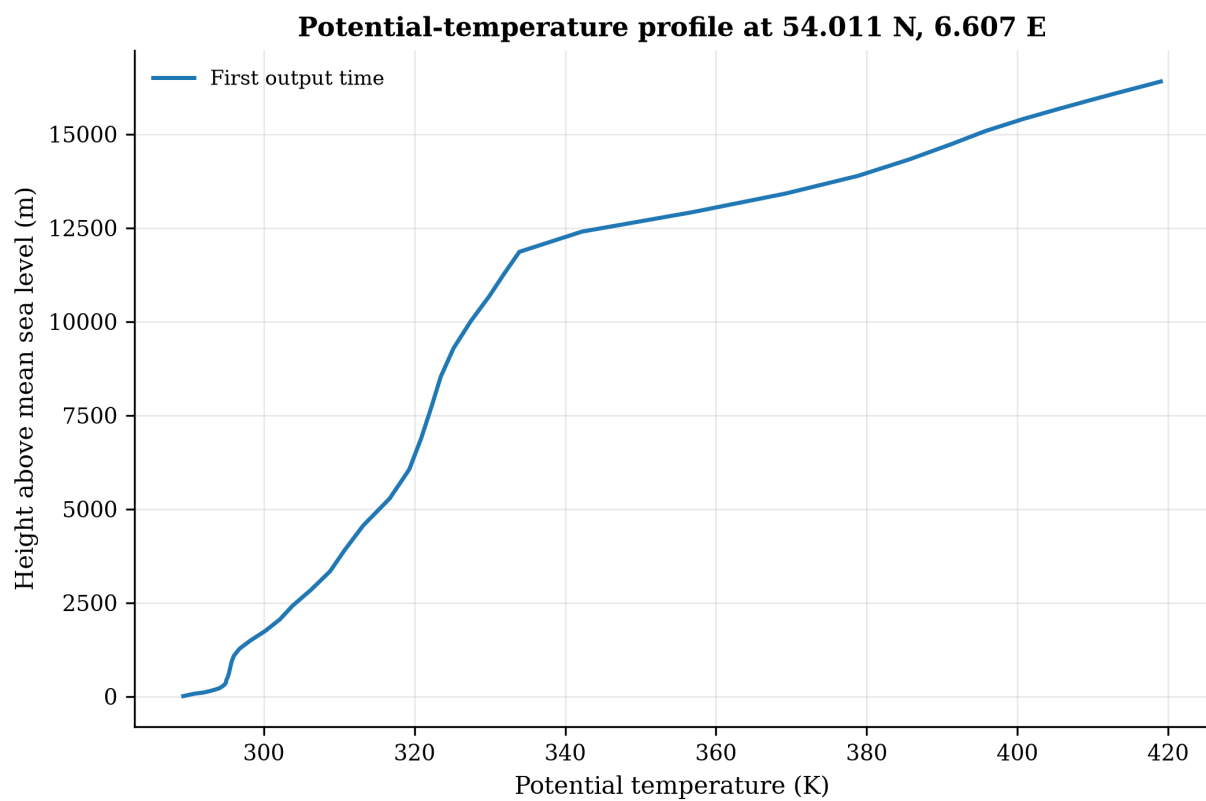
Diagnostic figures



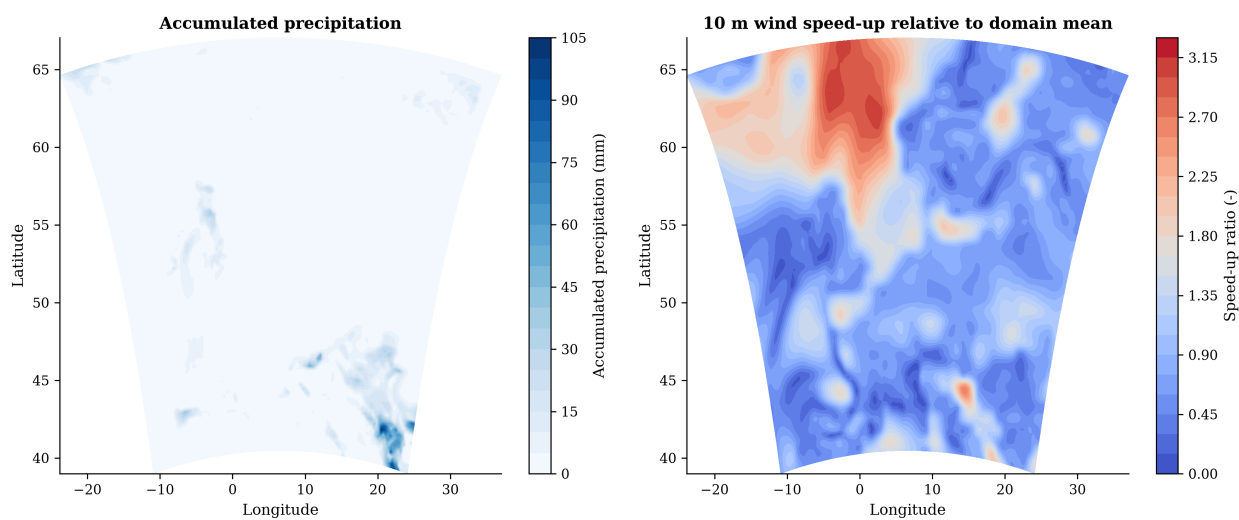
Boundary Layer Height



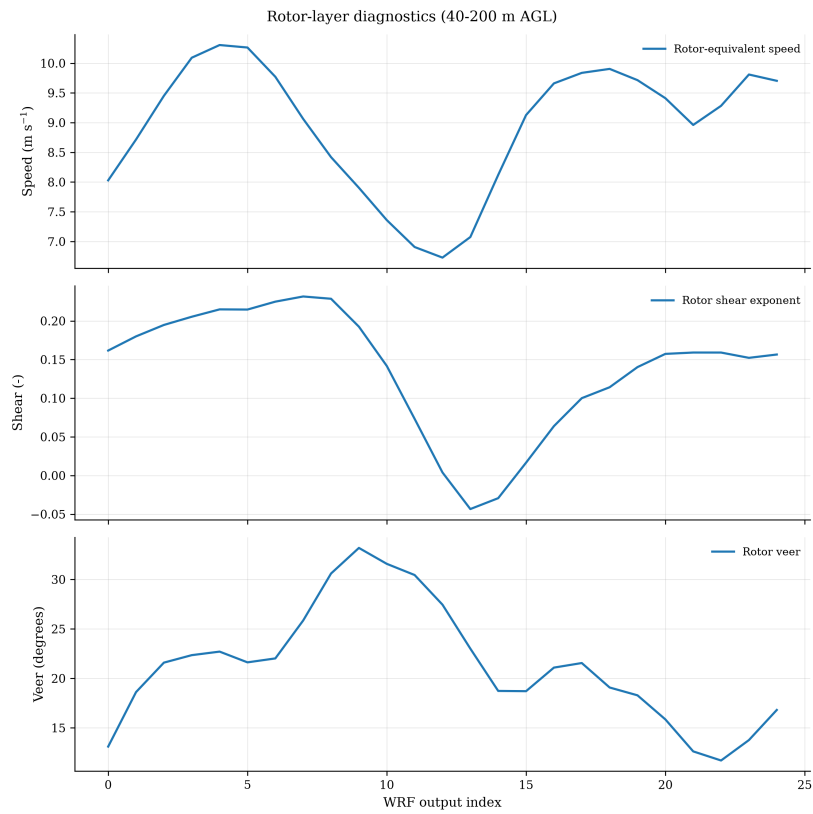
Flow Structure Diagnostics



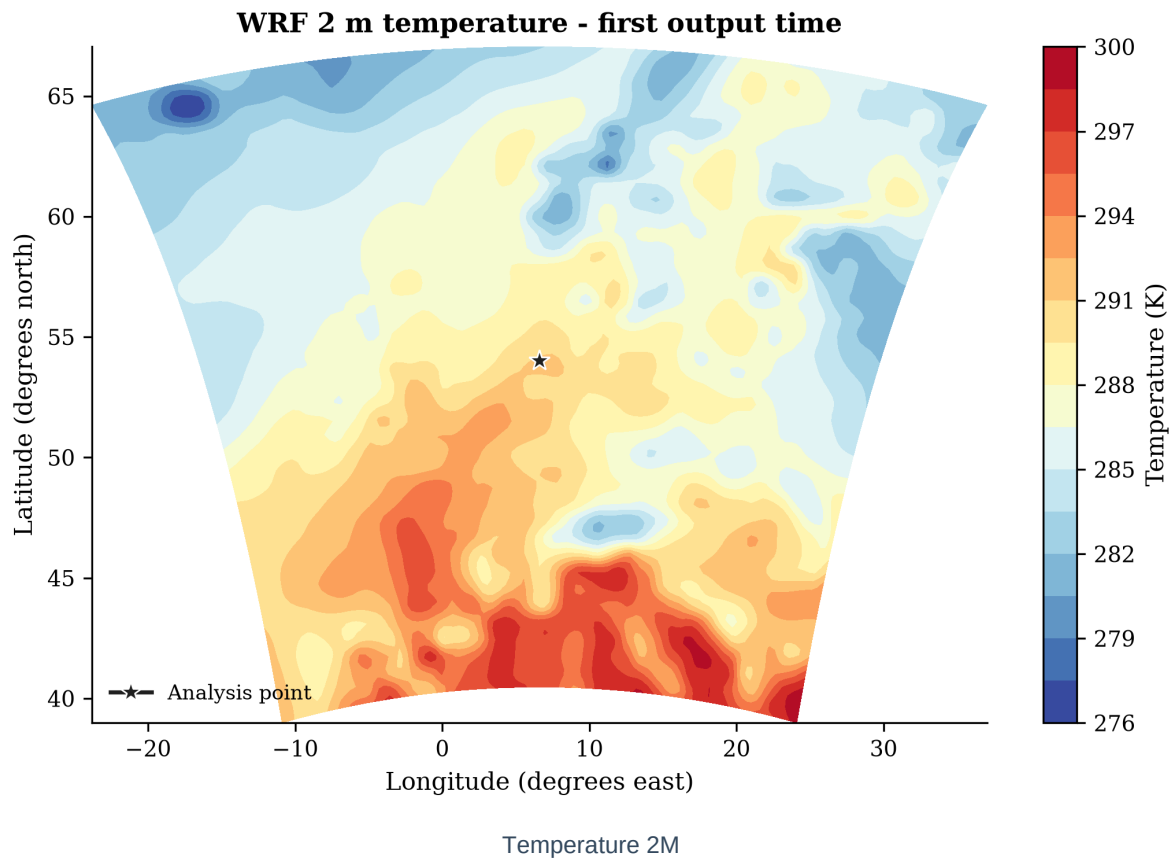
Potential Temperature Profile

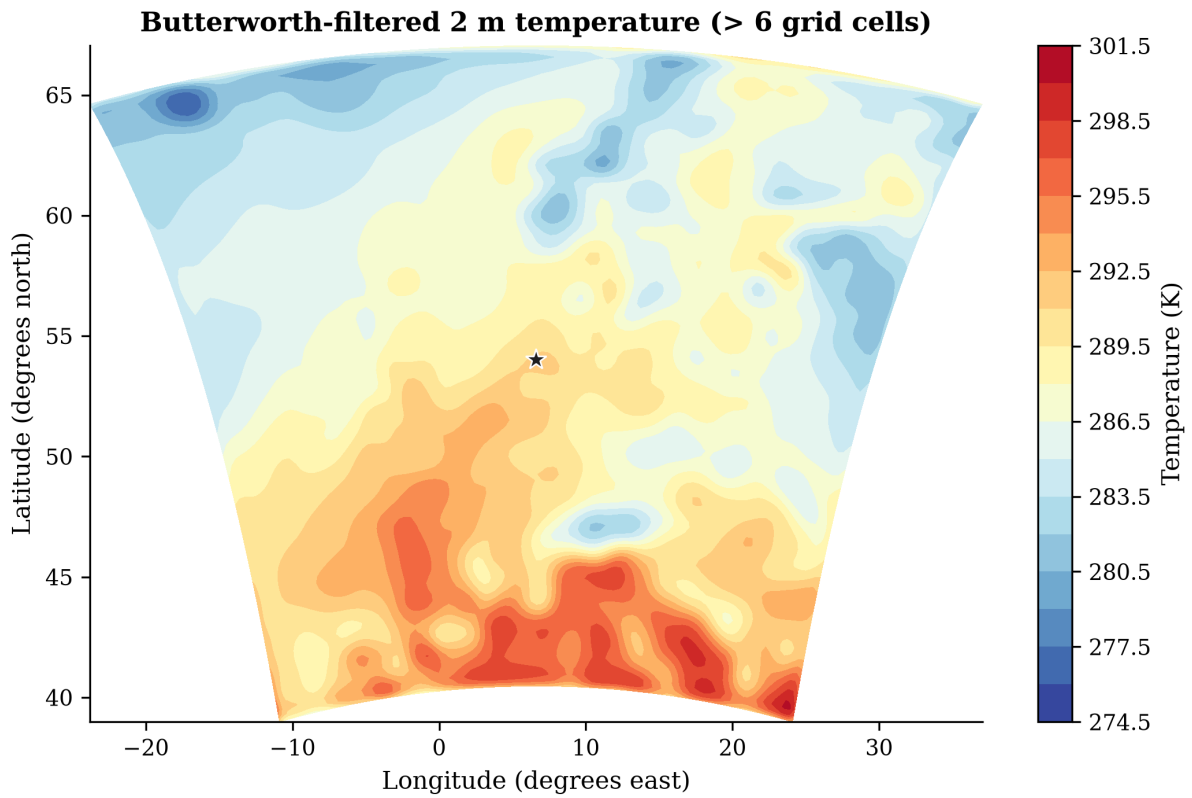


Precipitation And Speedup

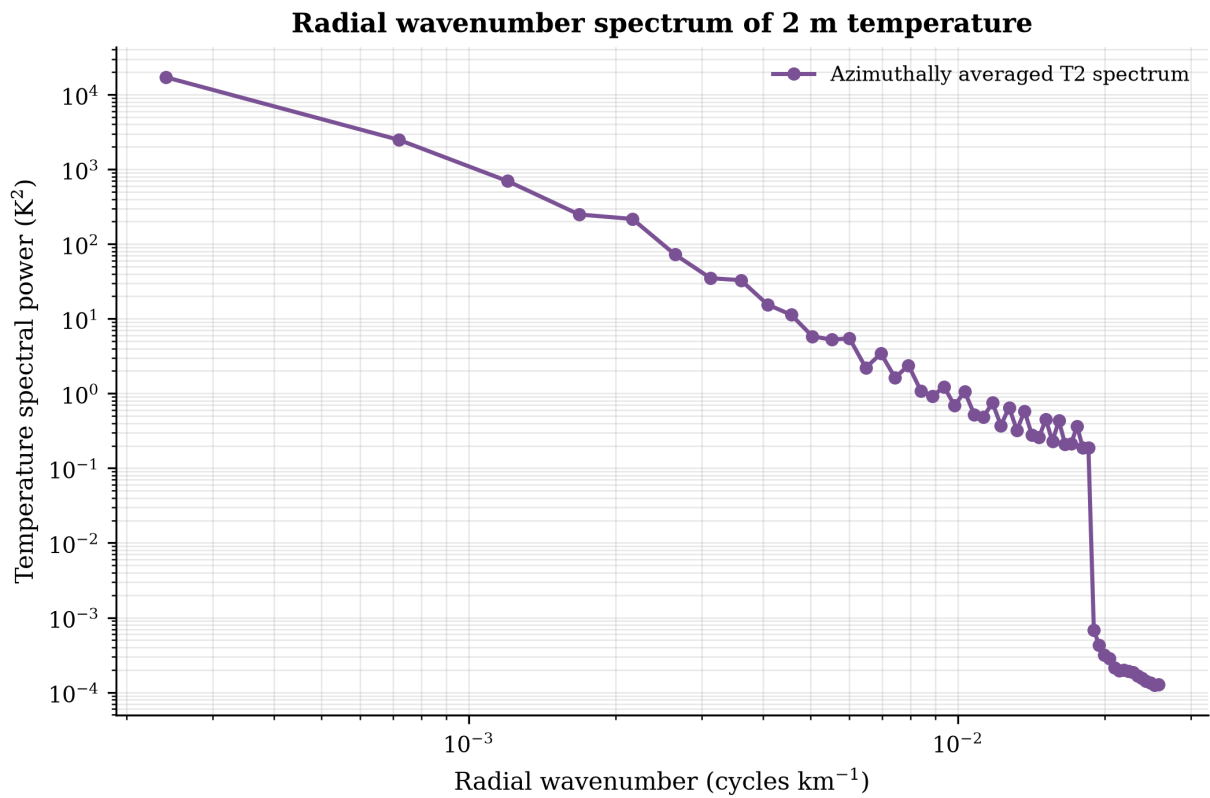


Rotor Diagnostics

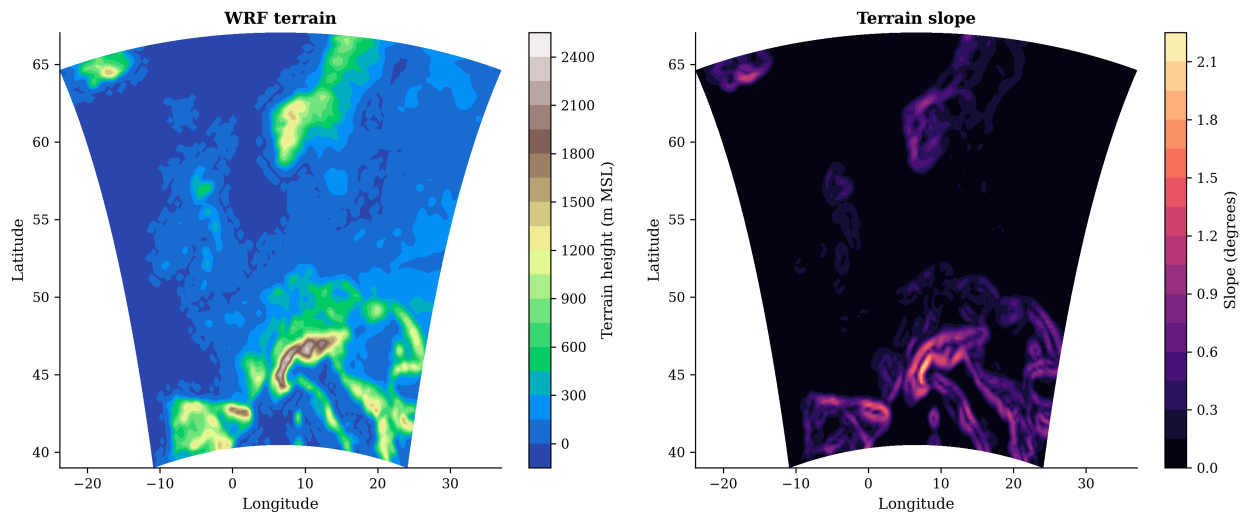




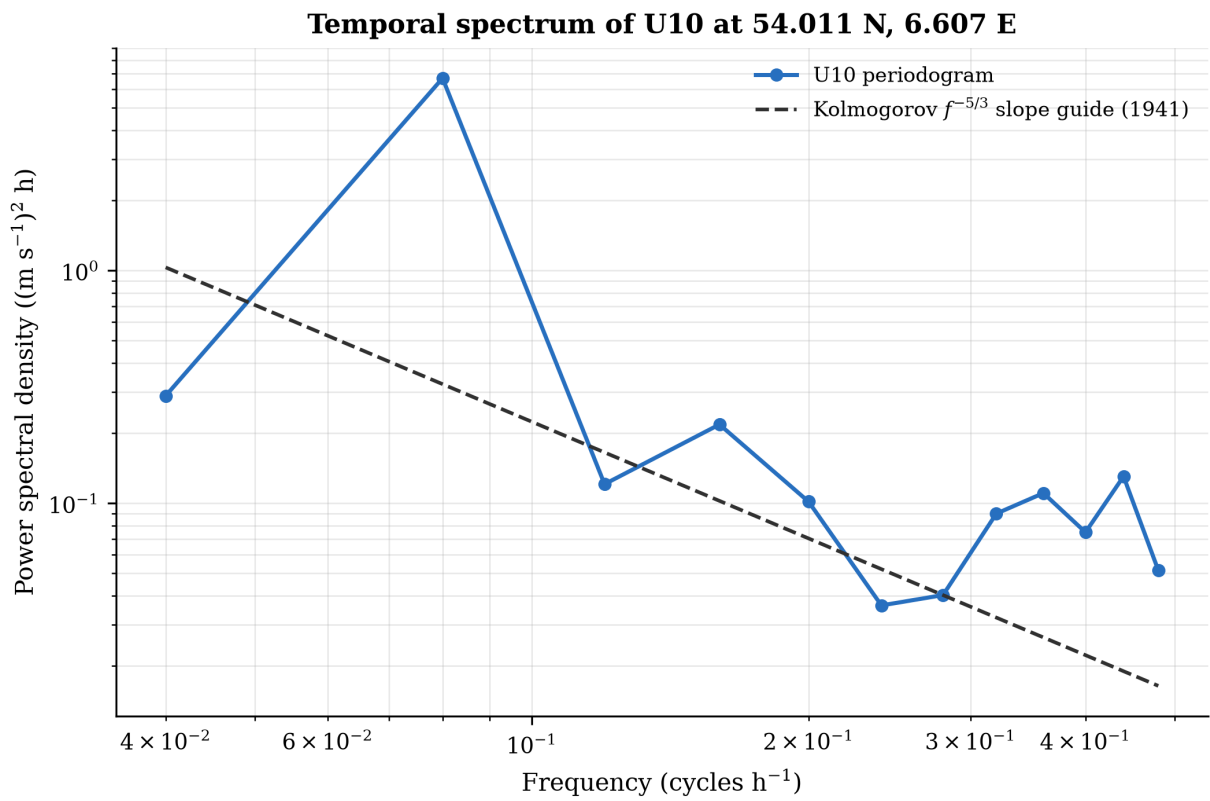
Temperature 2M Filtered



Temperature 2M Wavenumber Spectrum

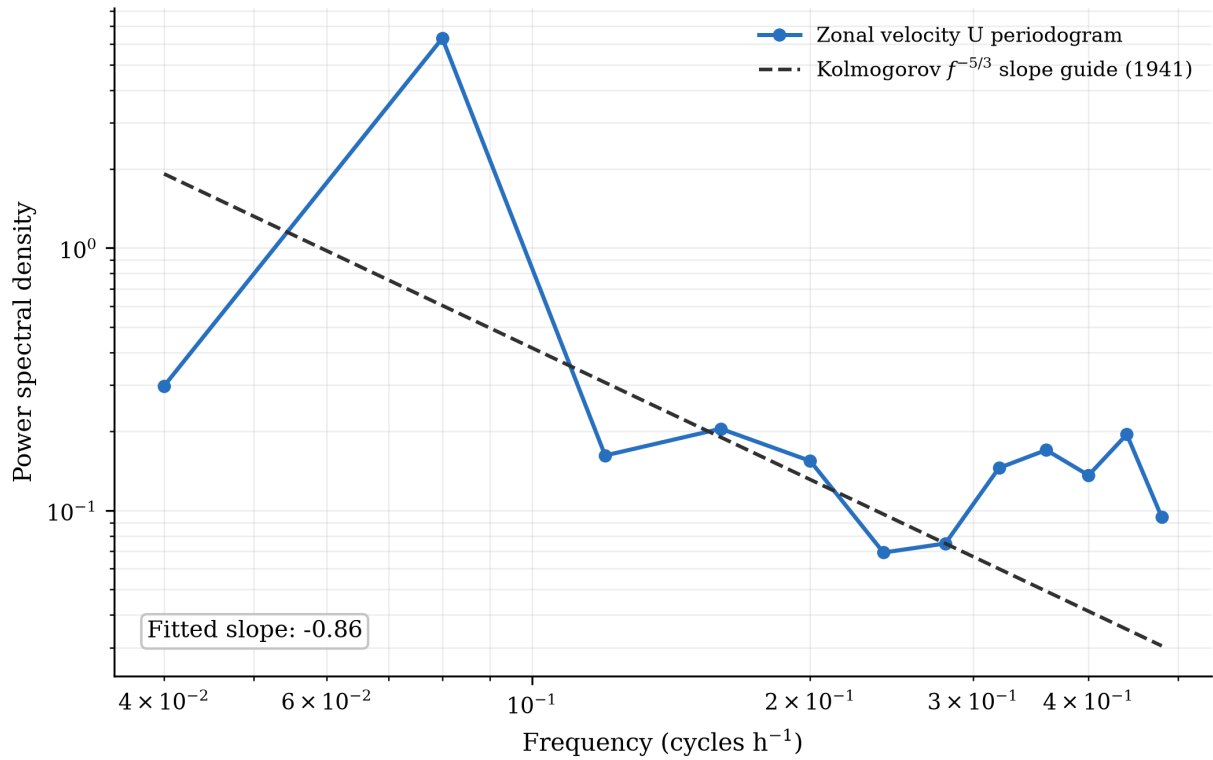


Terrain Diagnostics



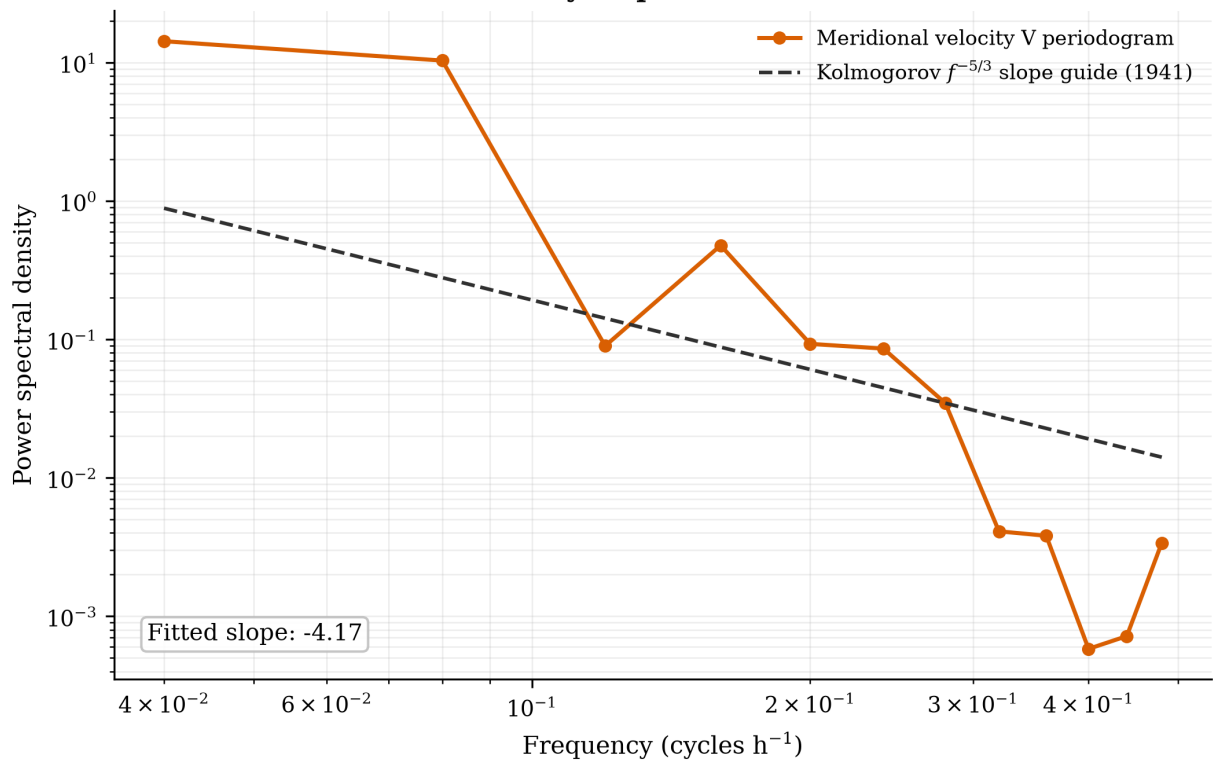
U10 Temporal Spectrum

Zonal velocity U spectrum - first model level

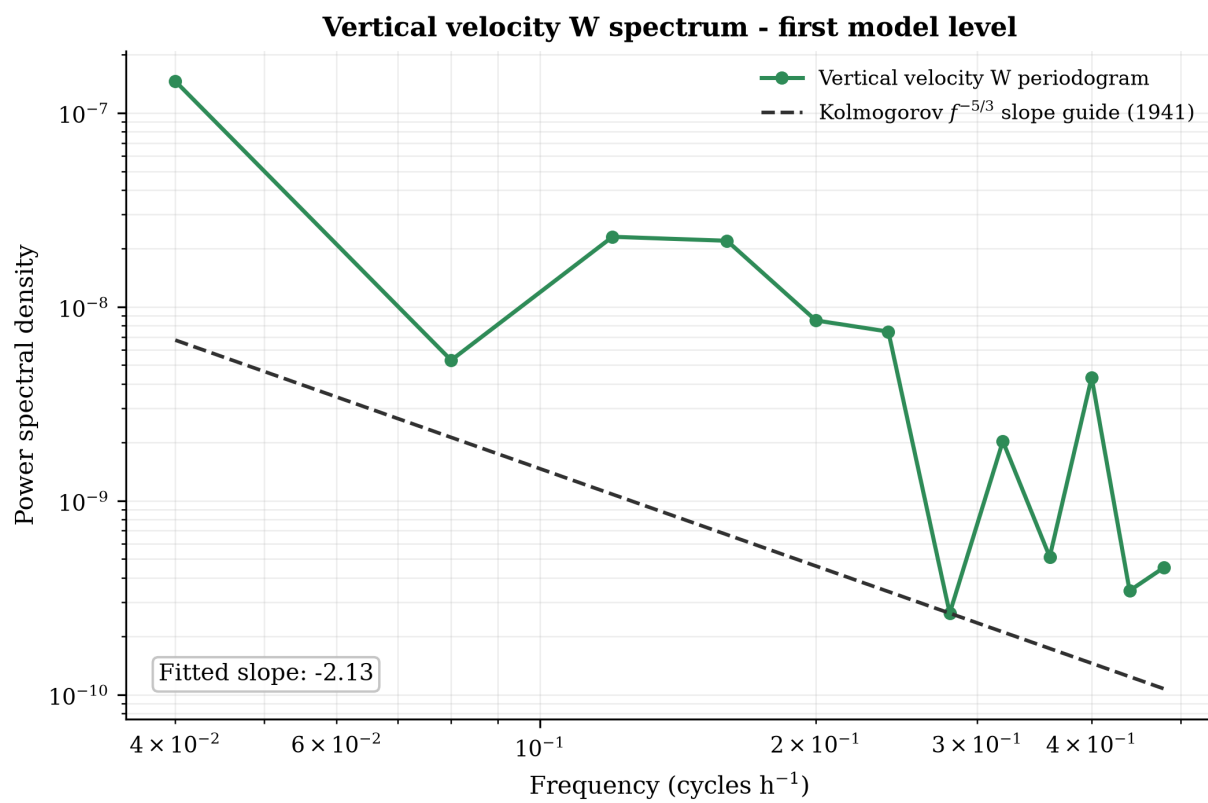


U Wind Spectrum

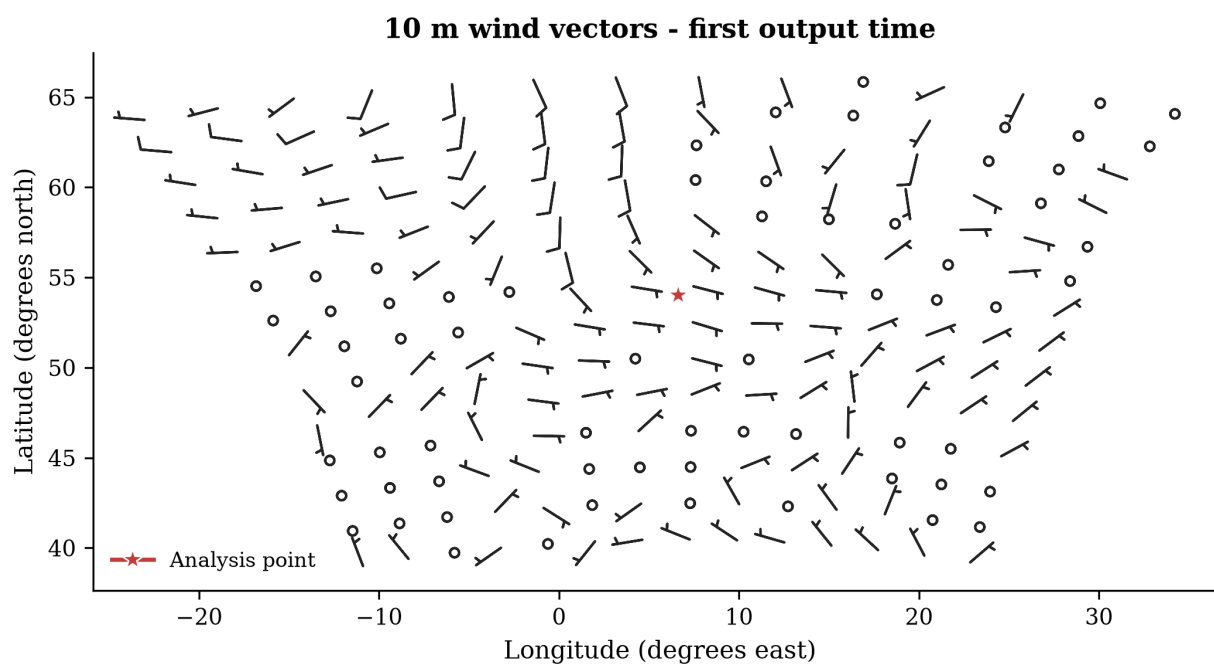
Meridional velocity V spectrum - first model level



V Wind Spectrum

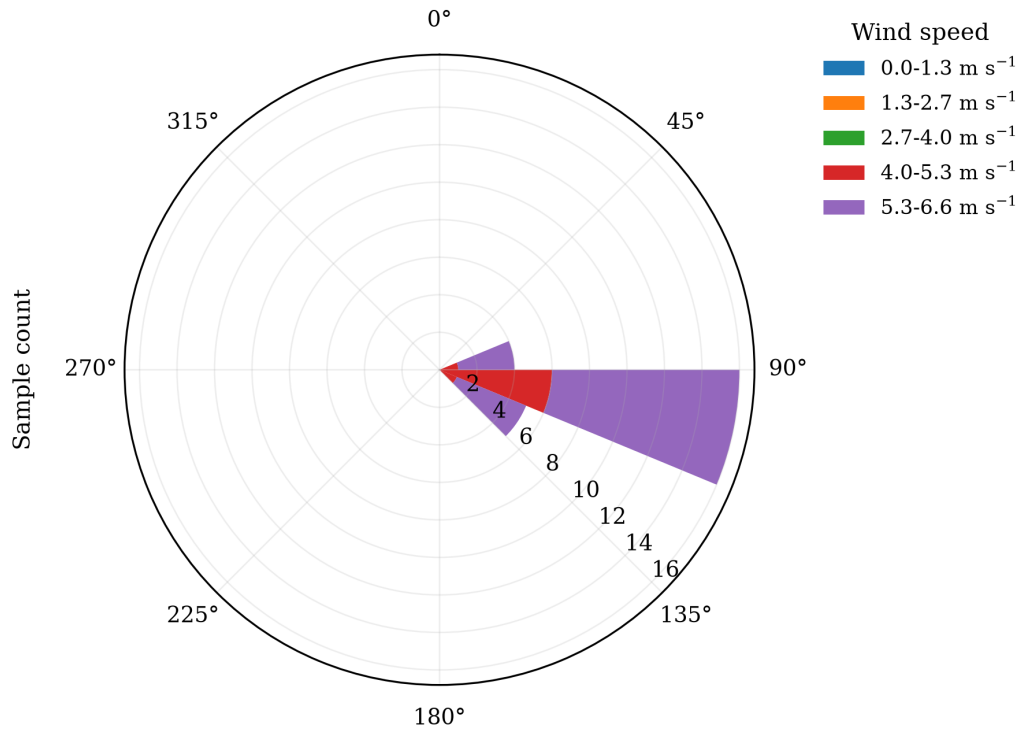


W Wind Spectrum



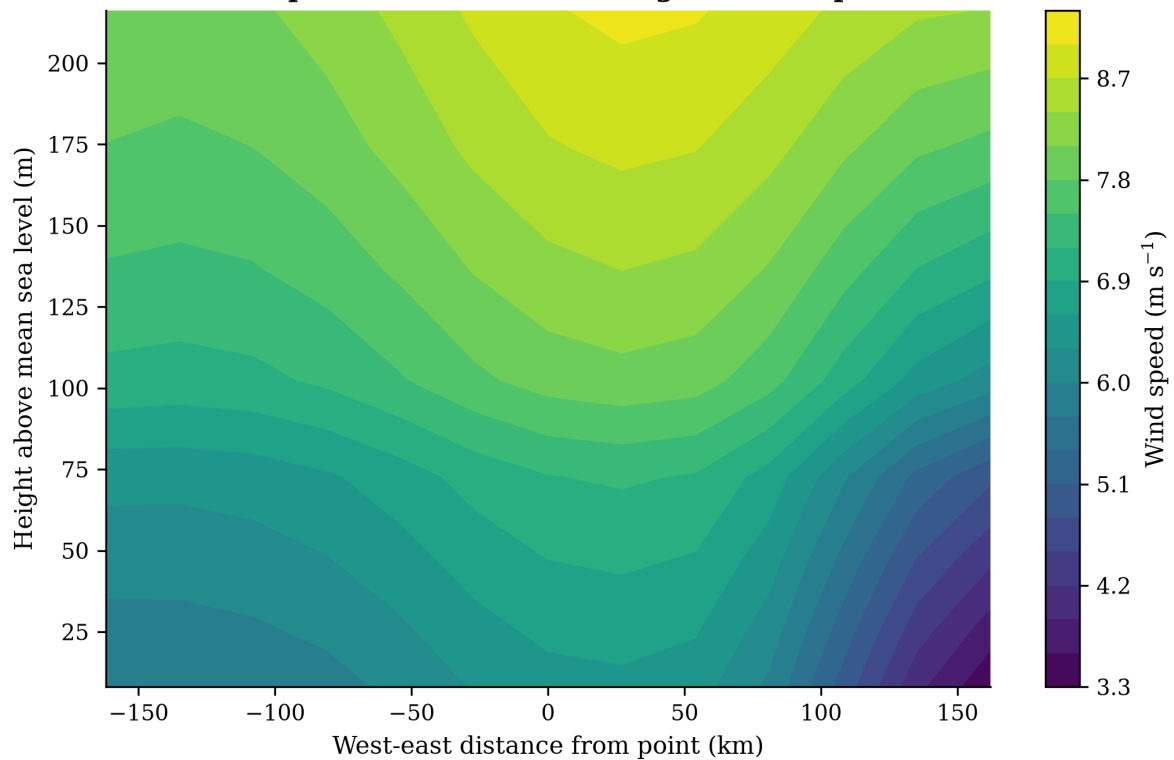
Wind Barbs 10M

10 m wind at selected grid cell



Wind Rose 10M

Wind-speed cross-section through selected point



Wind Speed Cross Section

